Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. When we initialize a Kalman filter state estimator, which of the following should we do? Select all that apply.

1 / 1 point

- [ ] We should always set all state estimates in $\hat{x}_0^+$ to zero.
- [x] We should always set all state estimates in $\hat{x}_0^+$ to our best guess of what those elements should be.

Correct
Yes. We should use all information that we have available to us to set the initial guess of $\hat{x}_0$.

- [ ] We should always set all elements of $\hat{x}_0^+$ to small zero-mean random values having covariance $\Sigma_{\tilde{x},0}^+$.
- [x] We should always set $\Sigma_{\tilde{x},0}^+$ to be large enough that the true initial state is within $\hat{x}_0^+ \pm 3\sqrt{\operatorname{diag}(\Sigma_{\tilde{x},0}^+)}$.

Correct
Yes. This is correct.

2. If we do not believe that the true initial state of the system x_0 is inside the error bounds $\hat{x}_0^+ \pm 3\sqrt{\operatorname{diag}(\Sigma_{\tilde{x},0}^+)}$, what should we do?

1 / 1 point

- ( ) We should increase $\Sigma_{\widetilde{w}}$.
- ( ) We should decrease $\Sigma_{\widetilde{w}}$.
- (•) We should increase $\Sigma_{\tilde{x},0}$.
- ( ) We should decrease $\Sigma_{\tilde{x},0}$.
- ( ) We should increase $\Sigma_{\tilde{v}}$.
- ( ) We should decrease $\Sigma_{\tilde{v}}$.

Correct
Yes. This is the correct approach to tuning the initial state error covariance.

3. If we believe that our linear Kalman filter state estimate is converging more slowly than it could, what should we try?

1 / 1 point

- ( ) We should increase $\Sigma_{\widetilde{w}}$.
- (•) We should decrease $\Sigma_{\widetilde{w}}$.
- ( ) We should increase $\Sigma_{\tilde{x},0}^+$.
- ( ) We should decrease $\Sigma_{\tilde{x},0}^+$.
- ( ) We should increase $\Sigma_{\tilde{v}}$.

Correct
Yes. This will cause the filter to believe that there is less process noise, leading to reduced state uncertainty in step 1b, and hence faster convergence.